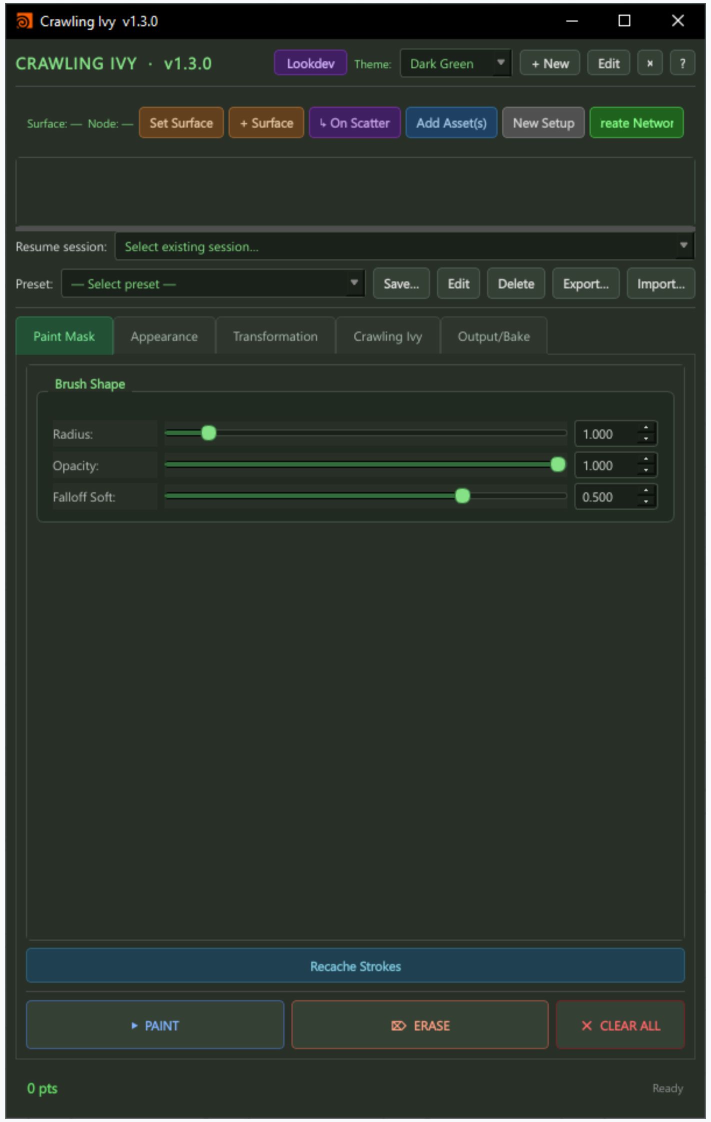
Crawling Ivy

v1.3.0 · Tutorial Guide

Surface-crawling vines for Houdini · 9 presets · Arnold · Redshift · Karma / USD

Paint Mask Tab · Define the seed region where crawling vines originate



Brush Shape

Radius — world-space brush size in the viewport

Opacity — stroke strength; lower values build density gradually

Falloff Soft — 0 = hard circular edge · 1 = smooth fade at brush perimeter

How to Use

Paint the starting zone — vines crawl outward from painted points

Denser painted areas spawn more crawling strands

The painted mask defines both density and origin of all vines

Use ERASE to trim back the mask region locally

CLEAR ALL resets the entire mask for this setup

Recache Strokes — force re-evaluation after changing brush parameters

Bottom Bar

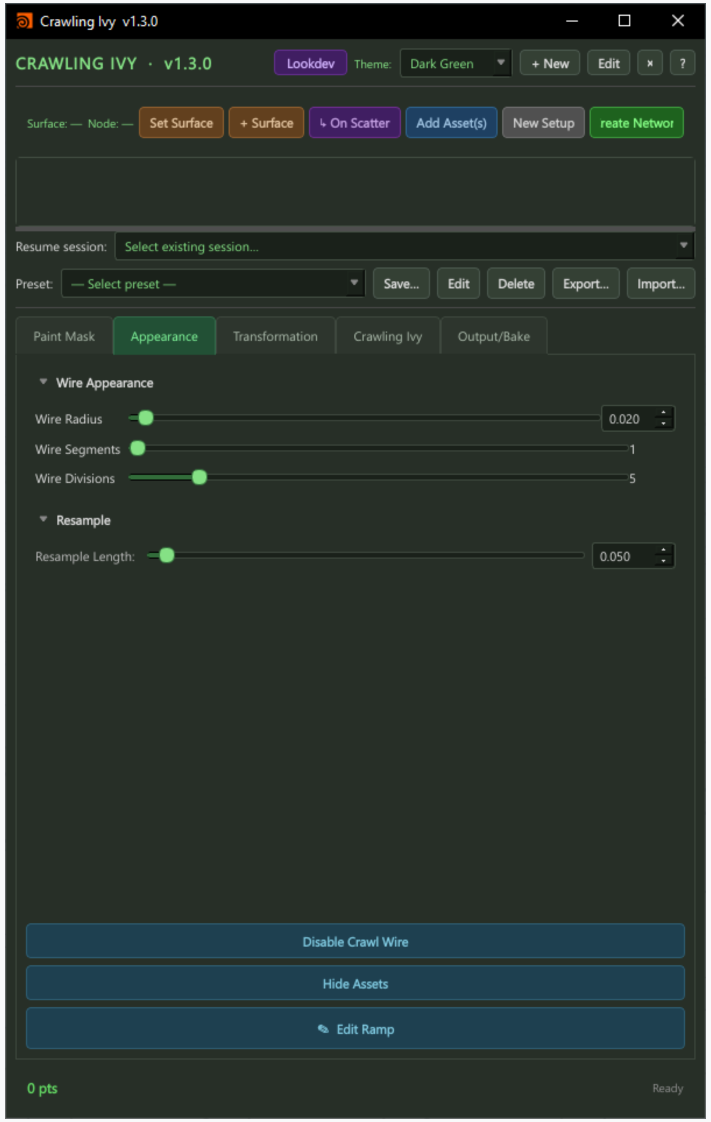
PAINT — activate brush; LMB drag across surface in the viewport

ERASE — remove painted mask in the brushed area

CLEAR ALL — wipe the entire density mask

Status line shows current point count and cook state

Appearance Tab · Wire thickness, geometry resolution, and resampling



Wire Appearance

Wire Radius — thickness of each crawling vine stem in world units
Wire Segments — polygon count around each wire cross-section (higher = rounder)
Wire Divisions — number of length-wise segments along each vine strand

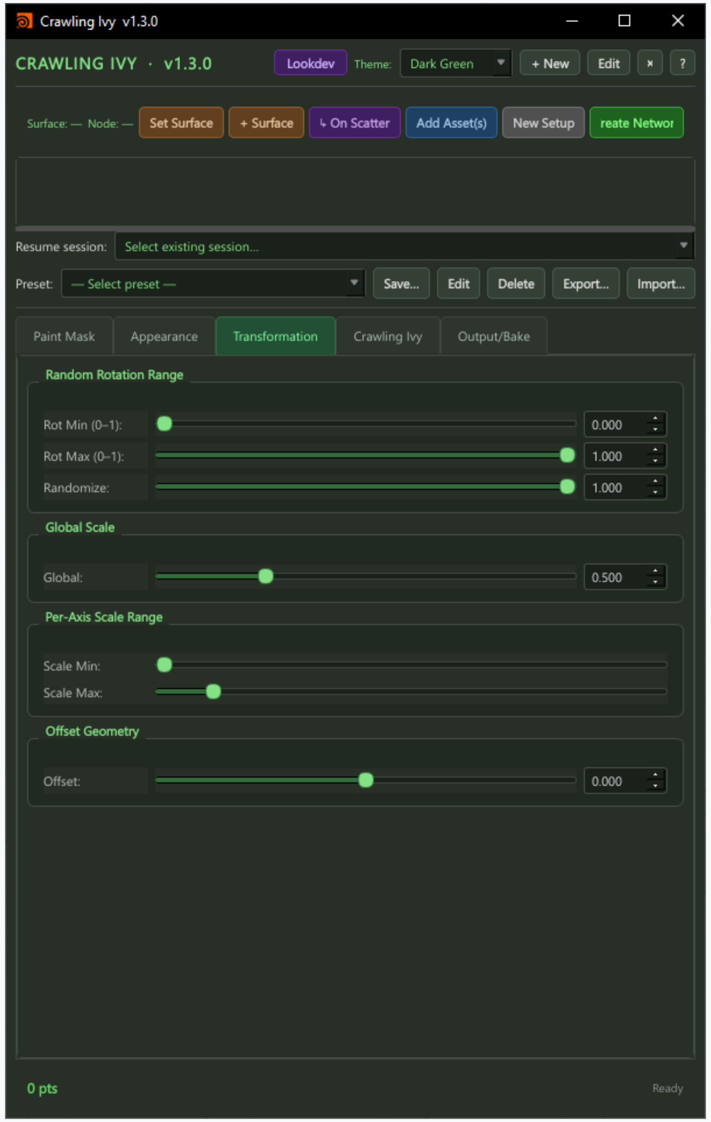
Resample

Resample Length — resamples the underlying crawl curve at this segment length
Lower values produce smoother, more detailed wire geometry
Higher values are faster to cook but produce chunkier wire curves

Action Buttons

Disable Crawl Wire — hides wire geometry in viewport for faster interaction
Hide Assets — hides leaf geometry; useful when shaping wire paths
Edit Ramp — opens the thickness-along-length ramp for tapering vine tips

Transformation Tab · Rotation and scale variation for crawling ivy leaf assets



Random Rotation Range

Rot Min (0–1) — minimum normalized rotation per leaf instance (0 = 0°)

Rot Max (0–1) — maximum normalized rotation per leaf instance (1 = 360°)

Randomize — blends between no variation and full Min/Max range

Global Scale

Global — uniform scale multiplier applied to every leaf asset instance

Use to size all leaves up or down without changing per-axis ranges

Per-Axis Scale Range

Scale Min — minimum random scale per leaf instance

Scale Max — maximum random scale per leaf instance

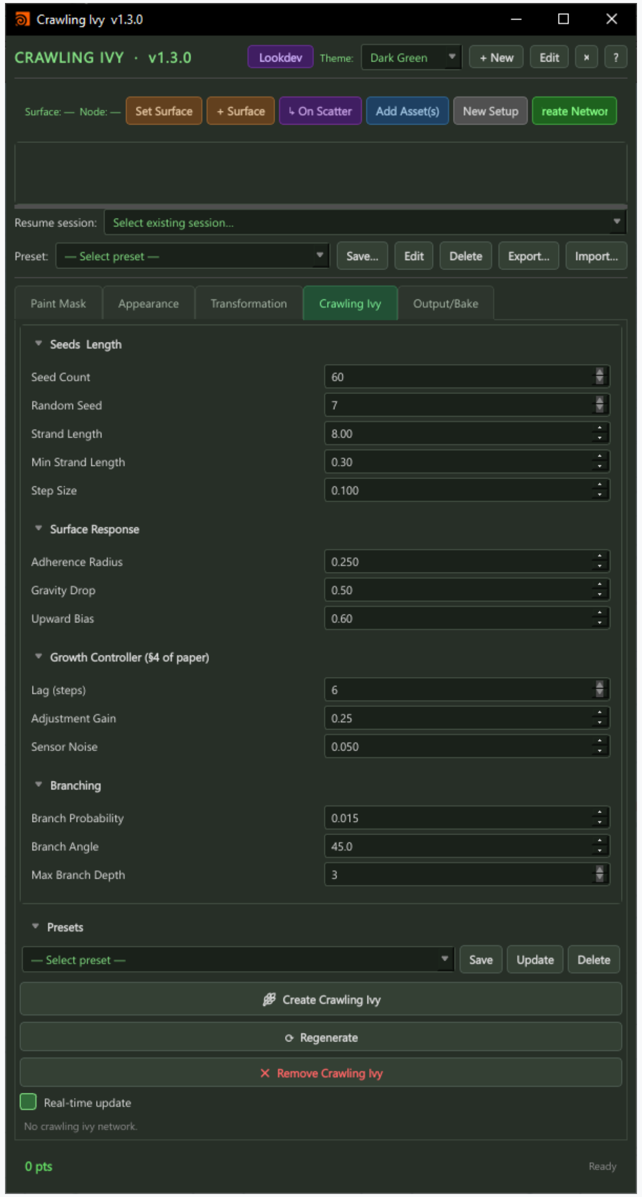
Each leaf gets a unique random scale between Min and Max

Offset Geometry

Offset — moves each leaf along its surface normal by this amount

Use to lift leaves off the wire surface to prevent clipping

Crawling Ivy Tab · Surface-crawling growth algorithm with branching control



Seeds / Length

Seed Count — number of individual crawling strands to generate
Random Seed — changes the placement pattern without altering other settings
Strand Length — maximum length each vine crawls before stopping
Min Strand Length — minimum length; strands shorter than this are culled
Step Size — distance advanced per growth step; smaller = smoother but slower

Surface Response

Adherence Radius — how closely strands follow the surface contour
Gravity Drop — downward pull as strands grow; higher = heavier droop
Upward Bias — counter-force pushing strands upward; balances gravity for climbing

Growth Controller (§4 of paper)

Lag (steps) — how many steps behind the strand direction adapts to the surface
Adjustment Gain — sensitivity of the direction correction each step
Sensor Noise — adds random variation to the surface sensing; more organic paths

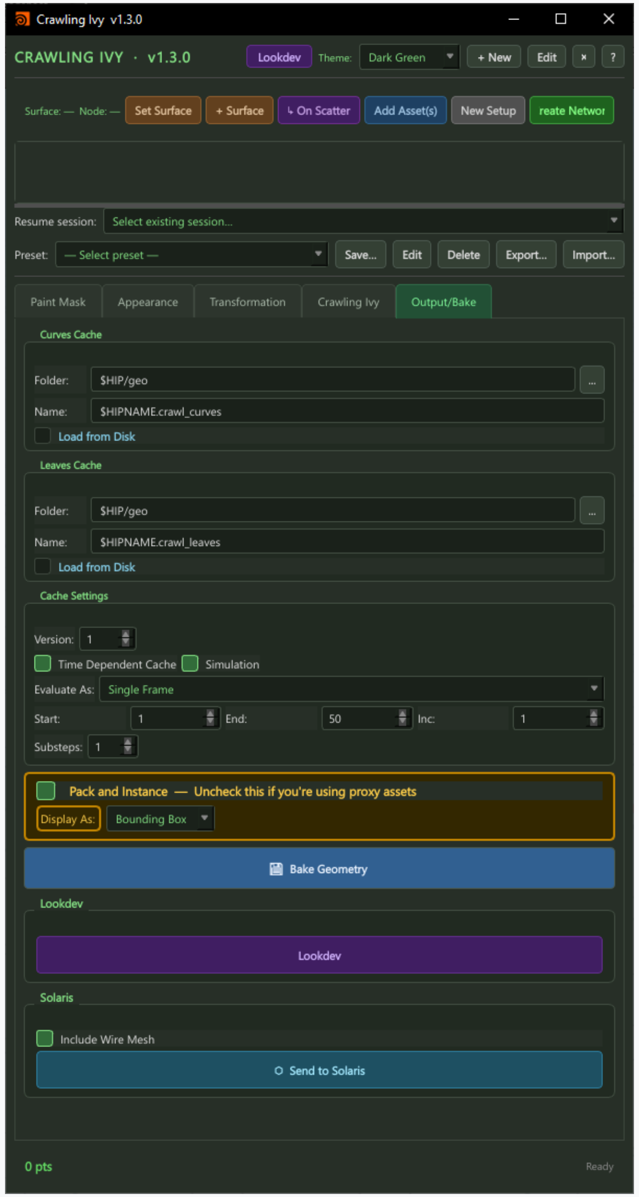
Branching

Branch Probability — chance per step of spawning a new sub-strand (0–1)
Branch Angle — angle in degrees between parent and child branch direction
Max Branch Depth — maximum nesting depth of branches (1 = no sub-branches)

Actions

Presets — save, load, or update named crawl configurations
Create Crawling Ivy — builds the Houdini SOP network
Regenerate — re-cooks with current parameters (no network rebuild needed)
Remove Crawling Ivy — destroys the generated network nodes
Real-time update — auto-regenerates on every parameter change

Output / Bake Tab · Separate curve and leaf caches, lookdev, and USD export



Curves Cache

Folder — output directory for vine curve geometry ($HIP/geo by default)
Name — filename template ($HIPNAME.crawl_curves)
Load from Disk — switch between live cook and cached curves

Leaves Cache

Separate cache track for leaf/asset geometry
Folder / Name — $HIP/geo · $HIPNAME.crawl_leaves
Load from Disk — load previously baked leaves independently of curves

Cache Settings

Version — increment to write a new version without overwriting previous bakes
Time Dependent Cache / Simulation — enable for animated or sim-driven crawl
Evaluate As — Single Frame or Frame Range for sequence baking
Start / End / Inc / Substeps — frame range and substep settings

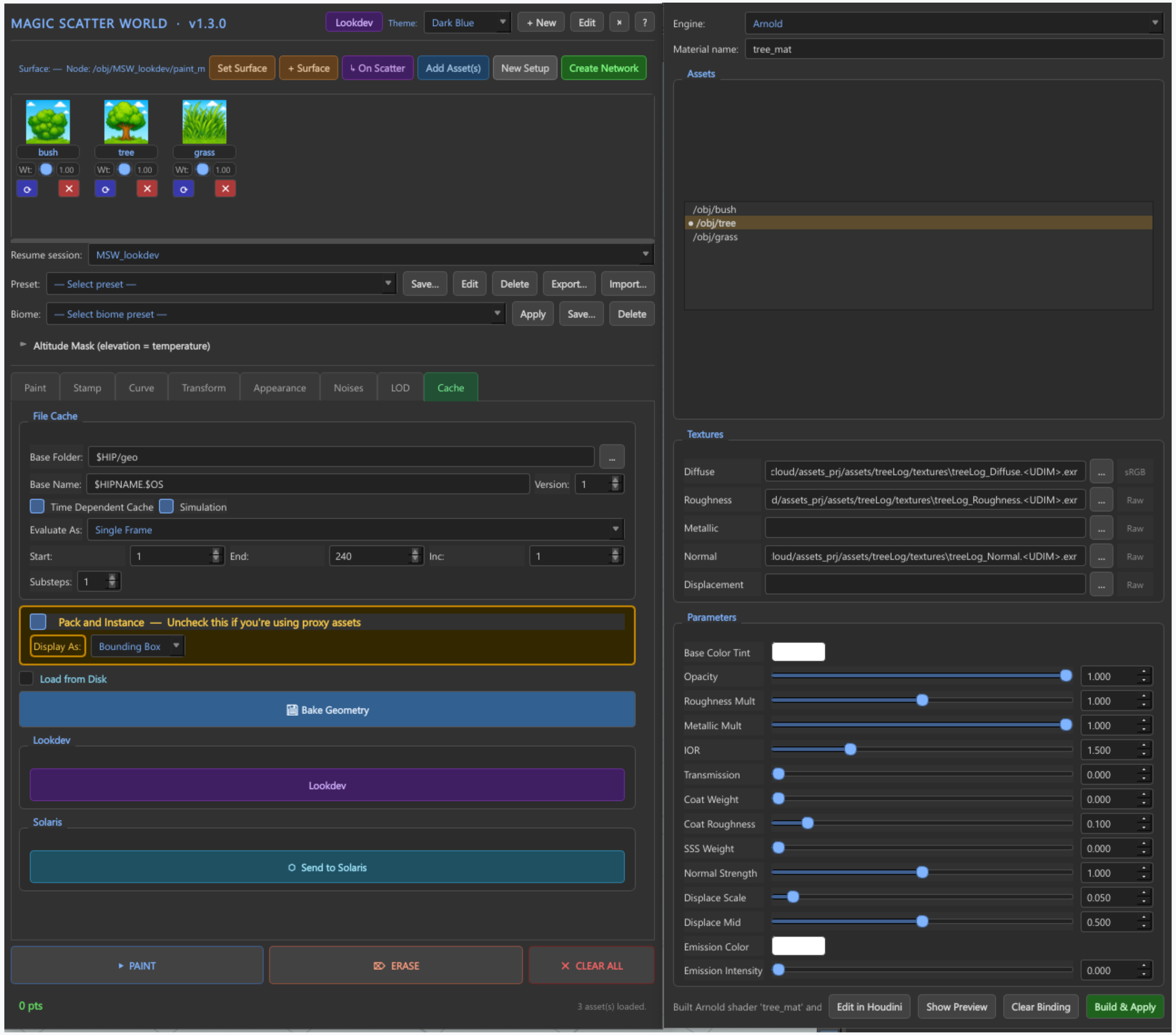
Ivy Baking

Pack and Instance — uncheck when using Redshift Proxy or Arnold Standin assets
Display As — Bounding Box (fast) or Full Geometry in viewport
Bake Geometry — freeze curves and leaves to disk at the paths above

Lookdev & Solaris

Lookdev — opens PBR material dialog (Arnold / Redshift / MaterialX)
Include Wire Mesh — include vine wire geometry in the USD export
Send to Solaris — exports as USD PointInstancer into a Solaris LOP network
Prototype USDs cached at $HIP/usd/prototypes/ with SHA1 invalidation

Lookdev · PBR material assignment · Arnold · Redshift · Karma / USD



Engine Selection

Arnold (HtoA) and Redshift native material network builders

Auto-detects the available renderer on tool launch

Texture Slots

5 slots: Diffuse · Roughness · Metallic · Normal · Displacement

Browse for any file path — supports UDIM glob patterns

Per-engine colorspace badges (sRGB / Raw) per slot

PBR Parameters

Roughness Mult — scale the roughness texture output

Metallic Mult — scale the metallic texture output

Normal Strength — intensity of the normal map

Coat Weight / Coat Roughness — clearcoat layer controls

SSS Weight — subsurface scattering amount

Emission Intensity / Emission Color — self-illumination

IOR — index of refraction for specular response

Transmission — glass and water transparency

Actions

Build & Bake — compile material network and assign to all assets

Show Painting — toggle asset visibility in viewport

Bake Geometry — freeze geometry with material assignments to disk

Send to Solaris — export with MaterialX shaders for Karma / USD

Quick Start Workflow · Grow crawling vines in minutes

1. Open from the Crawling Ivy shelf button
2. Select your surface geometry → click Set Surface
3. Select leaf/asset geometry → click Add Asset(s)
4. Click Create Network and name your setup
5. Paint Mask tab → paint the seed region where vines start
6. Crawling Ivy tab → click Create Crawling Ivy
7. Adjust Strand Length, Step Size, Adherence Radius
8. Tune Branching for density and spread
9. Appearance tab — set Wire Radius and Resample Length
10. Transformation tab — add scale and rotation variation to leaves
11. Output/Bake tab → click Bake Geometry to freeze the result
12. Click Send to Solaris for USD / Karma rendering

Enable Real-time update for instant feedback while tweaking parameters.